

Midterm - Results



Attempt 1 of 1

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Attempt Score **36 / 47 - 76.6 %**

Overall Grade (Highest Attempt) **36 / 47 - 76.6 %**

Question 1

1 / 1 point

In Naive Bayes, Laplace smoothing is used to normalize all features values between 0 and 1.

- ☐ True
- ✓ ☒ False

Question 2

1 / 1 point

Out of the following options, which is the least important part of a sentence that can be removed when encoding the sentence using bag-of-words?

- ☐ common nouns
- ✓ ☒ stopwords
- ☐ any word that occurs more than once in the input sentence
- ☐ proper nouns such as people's names, since they won't be present in the embedding scheme anyway

Question 3

1 / 1 point

Naive Bayes assumes that all features follow a Markov assumption (that is, every word depends upon the previous word in the sequence).

- ☐ True

✓ ☒ False

Question 4

0 / 1 point

Which of the following are true differences between linear models and neural networks? Choose all that apply.

- ✓ ☐ Linear models can use word vectors, while neural networks cannot.
- ✗ ☒ Linear models cannot be sensitive to the order of words in a sentence, while neural networks can be sensitive to word order.
- ✓ ☐ Linear models can only perform binary classification, while neural networks can also perform multi-class classification.
- ➔ ✓ ☒ Training linear models is typically a convex problem, while training neural networks is non-convex.

Question 5

1 / 1 point

The Word2Vec objective function closely resembles the loss function used in logistic regression. True or false: Training a Word2Vec model is a convex optimization problem because it relies on this logistic regression-like loss function.

- ☐ True
- ✓ ☒ False

Question 6

1 / 1 point

We want to create a parametric function that maps words from our vocabulary to vectors of length d . Suppose we have a vocabulary of 15000 unique words and we want to learn word vectors of dimension $d = 20$. How many total parameters will we learn? Please type in the exact number, and do not use any commas, full-stops, etc.

Answer: 300000 ✓

Question 7

1 / 1 point

For questions 7 & 8: Word vectors allow us to complete relations using vector arithmetic, forming parallelograms in the vector space. Suppose we want to solve the analogy query: "loud is to loudest as dark is to..."

Which of the following vector operations would best compute the target word v for this analogy?

- ☐ $v \approx v_{\text{loud}} - v_{\text{loudest}} + v_{\text{dark}}$
- ✓ ☒ $v \approx v_{\text{loudest}} - v_{\text{loud}} + v_{\text{dark}}$
- ☐ $v \approx v_{\text{loud}} + v_{\text{loudest}} + v_{\text{dark}}$
- ☐ $v \approx v_{\text{loud}} - v_{\text{dark}} + v_{\text{loudest}}$

Question 8

0 / 1 point

True or False: When searching the vocabulary for the word whose vector is most similar to our calculated vector v , we typically need to exclude words that are very similar to the query word itself.

- ➡ ☐ True
- ✗ ☒ False

Question 9

0 / 1 point

We explored two primary methods for capturing the Distributional Hypothesis: building raw co-occurrence matrices and training dense Word2Vec embeddings. Which of the following statements correctly compare these approaches? (Select all that apply)

- ➡ ✓ ☒ Word2Vec embeddings are explicitly trained to predict context words, while co-occurrence matrices only count them.
- ✓ ☐ Because they contain exact counts, raw co-occurrence vectors are generally dense, whereas Word2Vec vectors are highly sparse.
- ➡ ✓ ☒ Both methods successfully rely on the premise that words appearing in similar contexts share similar meanings.
- ➡ ✗ ☐ Raw co-occurrence vectors have a dimensionality directly tied to the size of the vocabulary.

Question 10

1 / 1 point

You implement a simple text classifier by applying mean pooling (averaging the word vectors) to represent entire sentences. Which of the following represents a strict mathematical or representational failure mode of this specific architecture? (Select all that apply)

- ☒ ☒ The sentence "birds eat worms" and the sentence "worms eat birds" will result in the exact same final document vector
- ☒ ☐ The model requires an extra projection layer because mean pooling a 10-word sentence creates a vector that is 10 times larger than the standard embedding dimension d.
- ☒ ☐ It prevents the classifier from leveraging word vectors trained on large, unannotated data to generalize to words not seen in the classifier's specific training data.
- ☒ ☒ If a document repeatedly uses the word "bank" to refer to a financial institution, and then uses "bank" once to refer to a river, the model cannot dynamically assign a different vector to the latter instance to resolve the polysemy.

Question 11

0 / 1 point

Case study, for questions 11-15: You are implementing a feed-forward neural network (FFN) to identify which language an input sentence is written in. The input is a sentence (ranging from 5-10 words), and the output is one of the following labels: English (label 0), French (label 1), German (label 2), Spanish (label 3). The FFN has an input layer, multiple hidden layers, and an output classification layer.

Which of these are mathematically valid methods to embed the input sentence into a vector? [you can pick one or more answers]

- ☒ ☒ Average the vectors of all the words.
- ☒ ☐ Pad each sentence to the maximum length of 10, then concatenate the vectors of all words. To represent the <pad> as a vector, use either a zero vector or a randomly generated vector.
- ☒ ☐ Simply concatenate the vectors of all the words.



Pad each sentence to the maximum length of 10, then concatenate the vectors of all words. Since some sentences will have multiple <pad> tokens at the end, use a different randomly generated vector for each <pad> token.

Question 12

1 / 1 point

Assume an average word embedding scheme for the input layer. Once you obtain the necessary word embeddings, which of the following options is the correct way to obtain the average? Note: The answer choices refer to “components” of a vector. If a vector is $[x_1, x_2, x_3]$, then x_1 , x_2 , and x_3 are the three components.

- ☐ First average all the components within each word vector, then concatenate these averages together to form a new vector to be used as the input embedding.
- ☐ Add the vectors, component-wise, then divide each component of the resultant vector by 10 (since 10 is the maximum number of words in the sentence).
- ☐ First sort the components of each vector into ascending order, then take the component-wise average across all vectors.
- ✓ ☒ Add the vectors, component-wise, then divide each component of the resultant vector by the number of words in the current sentence.

Question 13

1 / 1 point

The input embedding is calculated using the average scheme mentioned above. Out of 100, 193, 5, 1000, which of the input embedding sizes can be used?

- ☐ Only 100 or 1000. Since the maximum number of words is 10, the embedding size has to be a multiple of 10.
- ☐ Anything except 5. Since some sentences have more than 5 words, 5 is not a feasible embedding size.
- ✓ ☒ Any of these sizes (100, 193, 5, 1000) work.
- ☐ Only 100, it's a standardized embedding size used popularly, it's neither too big nor too small.

Question 14

1 / 1 point

You are building an FFN with 3 hidden layers. Let's assume that the input layer is of size 300, and the hidden layers are of sizes [10, 500, 500]. What is the size of the weight matrix between the third hidden layer and the output layer?

- ☐ 500 x 1000
- ☐ 500 x 1
- ☐ 500 x 10
- ✓ ☒ 500 x 4

Question 15

1 / 1 point

Let's use ReLU activation for all hidden layers. Assume that for some input datapoint, the input embedding is transformed into the vector [0.1, -0.05, 0.9, 0.0011, -0.09, -0.01, 1.8, 1.9, -0.00001, 0.0], before ReLU is applied (that is, this is the intermediate vector before we obtain the vector at the first hidden layer). After ReLU is applied, how many zeroes will this vector have?

- ☐ 1
- ☐ 0
- ☐ 4
- ✓ ☒ 5

Question 16

1 / 1 point

What is the difference between the sigmoid, tanh and ReLU activation functions?

- ✓ ☒ sigmoid and tanh produce values in the range [0,1] and [-1,1] respectively, whereas ReLU produces values in the range (0, infinity)
- ☐ sigmoid and tanh produce values in the range [-1,1] and [0,1] respectively, whereas ReLU produces values in the range (0, infinity)
- ☐ sigmoid and tanh produce values in the range [0,1] and [-1,0] respectively, whereas ReLU produces values in the range (-infinity,

infinity)

- ☐ sigmoid and tanh produce values in the range $[0,1]$ and $[-1,1]$ respectively, whereas ReLU produces values in the range $(-\infty, \infty)$

Question 17

1 / 1 point

Recall that a convolutional neural network (CNN) looks at a sliding window of words in a sentence. True or false: A CNN with a sliding window size of 1 is equivalent to a bag of words model.

- ✓ ☒ True
- ☐ False

Question 18

0 / 1 point

Sahana is training a neural network classifier, but the training loss of her model is not decreasing during training. Which of the following are reasonable things Sahana could try to fix this problem?

- ✓ ☒ Use an optimizer like Adam instead of vanilla SGD
- ✓ ☒ Decrease the learning rate
- ✓ ☐ Initialize all parameters to all zeroes
- ✗ ☐ Increase the learning rate

Question 19

1 / 1 point

A Simple Recurrent Neural Network (RNN) computes the hidden state h_t by applying a non-linear function (like tanh) to the sum of the current input x_t and the previous hidden state h_{t-1} , multiplied by their respective weight matrices. What is the primary reason why this simple architecture struggles to model long-distance dependencies in text?

- ☐ The model evaluates the entire sequence simultaneously, running out of memory.
- ☐ Simple RNNs do not use non-linear activation functions.

- ✓ ☒ The repeated multiplication by the hidden state weight matrix during backpropagation causes gradients to vanish or explode.
- ☐ The word embeddings cannot capture semantic meaning over long sequences.

Question 20

0 / 1 point

Which of the following statements accurately describe the advantages or characteristics of a Bidirectional RNN (or Bi-LSTM) compared to a standard unidirectional RNN for sequence labeling? (Select all that apply)

- ✓ ☐ Bidirectional models require less memory to train because they process half the sequence at a time.
- ✗ ☐ Bidirectional models concatenate a forward hidden state and a backward hidden state to form the final representation at a given time step.
- ✓ ☒ Bidirectional models are highly effective for sequence labeling tasks, provided the entire input sequence is accessible at once.
- ✓ ☒ Bidirectional models can utilize future contextual information (right context) when making a prediction for a current word.

Question 21

1 / 1 point

For questions 21-22: Consider a Long Short-Term Memory (LSTM) cell processing a sequence of words. At time step t , the model encounters a highly irrelevant filler word (like "um" or "uh") that carries no useful information for the future context. However, the previous cell state c_{t-1} contains vital long-term subject-verb agreement information that must be preserved exactly as it is for future time steps. For reference, LSTM cell state update relies on the following equation:

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

, where c_{t-1} is the previous cell state, f_t is the forget gate, i_t is the input gate, and $\tilde{c}_t$ is the candidate cell state.

To achieve this exact preservation of the previous cell state (c_{t-1}) without adding any new information from the current filler word, what ideal values should the forget gate (f_t) and the input gate (i_t) approximate?

- ☐ $f_t \approx 0, i_t \approx 1$
- ☐ $f_t \approx 1, i_t \approx 1$
- ☐ $f_t \approx 0, i_t \approx 0$
- ✓ ☒ $f_t \approx 1, i_t \approx 0$

Question 22

1 / 1 point

True or False: If the output gate (o_t) evaluates to a vector of zeros, the final hidden state (h_t) emitted by the cell at time t will also be a vector of zeros, even though the internal cell memory (c_t) successfully preserved the vital long-term information.

- ✓ ☒ True
- ☐ False

Question 23

1 / 1 point

In the BLSTM-CNN architecture for sequence labeling, what specific limitation of the standard word-level BLSTM does the Convolutional Neural Network (CNN) layer address?

- ☐ It speeds up the sequential computation of the recurrent layers.
- ☐ It prevents the exploding gradient problem in the LSTM memory cell.
- ☐ It globally normalizes the output probabilities of the tags.
- ✓ ☒ It models character-level and spelling features that are important for identifying parts of speech.

Question 24

1 / 1 point

When using a Conditional Random Field (CRF) for sequence labeling, evaluating the normalization term over all possible output sequences directly is computationally intractable due to the exponential output space.

Which algorithm is used to efficiently find the most probable sequence of tags?

- ☐ Convolutional Max Pooling
- ✓ ☒ The Viterbi Algorithm
- ☐ The Most Frequent Class Baseline
- ☐ Backpropagation Through Time

Question 25

1 / 1 point

In Statistical Machine Translation (SMT), IBM Models utilize a word-level alignment model. Suppose you are translating a short foreign sentence containing 3 words into an English sentence containing 4 words. Assuming each foreign word can be aligned to any of the English words or to a special "Null" token, exactly how many possible alignments exist for this sentence pair? Please give an exact number as the answer, with no commas, full-stops, etc.

Answer: 125 ✓

Question 26

1 / 1 point

How does sequence-to-sequence (Seq2seq) generation fundamentally differ from sequence labeling tasks like Part-of-Speech tagging? (Select all that apply)

- ✓ ☐ In Seq2seq generation, the length of the output sequence must strictly match the length of the input sequence.
- ✓ ☐ Autoregressive factorization explicitly assumes statistical independence between all generated output tokens.
- ✓ ☒ The output space in Seq2seq generation is often much larger than in sequence labeling.
- ✓ ☒ Autoregressive Seq2seq models factorize the probability distribution by predicting the next token conditioned on the history of previously generated tokens.

Question 27

1 / 1 point

You are using Beam Search with a beam size of $K=2$. At step $t=1$, your top hypothesis is the word "he" with a cumulative log probability score of -0.7. At step $t=2$, the model evaluates the next possible words. The transition log probability of generating "hit" given "he" is -1.0. What is the new cumulative score for the partial hypothesis "he hit"? Please give a numerical answer. Commas are not allowed, but minus sign and decimal point are allowed if needed.

Answer: -1.7 ✓

Question 28

1 / 1 point

True or False: A major issue with the vanilla Encoder-Decoder architecture (without attention) is that a single encoding vector must capture all necessary information about the source sentence, which creates a bottleneck and can lead to vanishing gradients for longer sequences.

- ✓ ☒ True
☐ False

Question 29

0 / 1 point

Which of the following is true for residual connections in a transformer?
One or more options can be correct.

- ✓ ☐ They serve as a constant, and do not affect the backpropagation gradient flow
✓ ☐ They are implemented by multiplying two vectors together
➔ ✓ ☒ They add to the backpropagation gradient flow
✗ ☒ Every layer that has a residual connection adds a model capacity equivalent to the embedding size at that layer

Question 30

1 / 1 point

True or False: In a feedforward network, if a neuron is "dead" (its pre-activation is always negative) under ReLU, its corresponding weights will receive zero gradient for all training samples

- ✓ ☒ True

☐ False

Question 31

1 / 1 point

In reverse-mode automatic differentiation, why is it necessary to store the activations calculated during the forward pass?

- ☐ They act as a regularization term
- ☐ They are used to calculate the Hessian matrix
- ✓ ☒ They are required to compute the local gradients during the backward pass
- ☐ They serve as the starting point for the next forward pass

Question 32

0 / 1 point

In a standard MLP with L layers, let N be the total number of weights and M be the number of neurons. Which of the following statements regarding the computational complexity of a single backpropagation pass (one forward and one backward pass) is True?

- ➔ ☐ The time complexity of computing gradients for all weights is $O(N)$, which is the same order of magnitude as the forward pass.
- ☐ The memory complexity is $O(L)$ because we only need to store the activations of the current layer being processed.
- ✗ ☒ The backward pass is significantly more expensive than the forward pass, scaling as $O(N^2)$
- ☐ Calculating the gradient with respect to the input requires a separate forward pass for every input dimension.

Question 33

1 / 1 point

(Transformers) How does "cross-attention" in the decoder differ from "self-attention"?

- ✓ ☒ Queries come from the decoder, while keys and values come from the encoder output
- ☐ Queries come from the encoder output, while keys and values come from the decoder.

- ☐ It is only used during pre-training
- ☐ It does not use the $\frac{1}{\sqrt{d_k}}$ scaling factor

Question 34

0 / 1 point

Which option best describes what happens if attention scores are all equal for a query (after masking)?

- ☐ Output equals the query vector
- ☒ Output becomes one-hot on the max key
- ☐ Output becomes zero vector
- ☐ Output becomes the mean of value vectors over allowed positions

Question 35

1 / 1 point

In a Multi-Query Attention (MQA) setup, what is the specific reduction in memory traffic compared to Multi-Head Attention (MHA)?

- ☒ It shares a single key and value head across all query heads.
- ☐ It reduces the embedding dimension by a factor of H (number of heads).
- ☐ It eliminates the feed-forward network.
- ☐ It uses only one query head for all key/value pairs.

Question 36

1 / 1 point

How does the transformer architecture encode the order of the tokens in the input sequence?

- ☐ The transformer has a recurrent layer that handles the bidirectional (left-to-right and right-to-left) order of the tokens
- ☐ The transformer has a recurrent layer that handles the unidirectional order of the tokens
- ☒ The transformer has to use explicit positional embeddings (for example, sinusoidal, rotary, etc.)

- ☐ The self-attention mechanism in the transformer handles the positional encoding

Question 37

1 / 1 point

Which vectors are multiplied to get the attention score between two tokens in the self-attention mechanism?

- ☐ Key and Value
- ☐ Query, Value and Key
- ☐ Query and Value
- ✓ ☒ Query and Key

Question 38

1 / 1 point

What is being masked in the decoder when a transformer is being trained?

- ☐ Padding tokens, since they don't contribute anything
- ☐ The end-of-sentence token and the padding tokens
- ✓ ☒ The future tokens in the output sequence
- ☐ All the unnecessary tokens from the input sequence, as decided by the attention scores

Question 39

1 / 1 point

In standard next-token language modeling, the training objective is:

- ☐ Minimize MSE between hidden states
- ☐ Minimize KL between model and data in embedding space
- ☐ Maximize mutual information between layers
- ✓ ☒ Maximize log-likelihood of the next token given previous tokens (cross-entropy)

Question 40

1 / 1 point

As the sampling "Temperature" approaches infinity, what happens to the output distribution?

- ☐ The model stops generating
- ✓ ☒ It becomes uniform (random)
- ☐ It becomes deterministic (greedy)
- ☐ The model only outputs the most common word

Question 41

1 / 1 point

True or False: Increasing context length at inference always improves factual accuracy and never increases hallucinations.

- ☐ True
- ✓ ☒ False

Question 42

0 / 1 point

For BERT pre-training, what is the main purpose of adding [MASK] tokens?

- ☐ Reduces the discrepancy between pre-training and fine-tuning
- ✗ ☒ Prevents the model from "cheating" by looking at future words
- ➔ ☐ Hides some words that the model has to guess
- ☐ Improves the efficiency of training

Question 43

0 / 1 point

Which of the following statements are true about Byte Pair Encoding?
Choose all that apply.

- ✗ ☒ It leads to longer token sequences compared to using a character-level vocabulary
- ✓ ☐ It is only used for multilingual tasks
- ➔ ✓ ☒ It requires use of a text dataset to create the vocabulary
- ➔ ✓ ☒ It helps dramatically with the problem of unknown words

Question 44

1 / 1 point

Which statement best describes Fully Sharded Data Parallel (FSDP) as presented in the lecture?

- ✓ ☒ Split each Transformer block's parameters and gather only the layer's parameters when computing that layer
- ☐ Partition hidden states across GPUs but replicate all parameters
- ☐ Split the training data only; keep full model parameters on every GPU
- ☐ Only shard optimizer states; keep parameters and gradients replicated

Question 45

1 / 1 point

Which communication pattern matches All-gather?

- ☐ Each GPU sums tensors and keeps only its local partial sum
- ✓ ☒ Each GPU starts with a shard; after operation, every GPU has the concatenation of all shards
- ☐ Each GPU broadcasts its shard to CPU only
- ☐ Each GPU starts with full tensor; after operation, each GPU keeps only one shard

Question 46

1 / 1 point

According to the "Three Criteria" for distributed pretraining, which of the following is NOT a primary goal?

- ✓ ☒ Maximum number of GPU nodes
- ☐ Minimal redundant computation
- ☐ Minimal communication overhead
- ☐ Minimal peak memory cost

Question 47

1 / 1 point

Which of the following correctly describes the mathematical relationship between basic GPU communication operations?

- ☐ All-gather = Reduce-scatter + All-reduce
- ☐ Reduce-scatter = All-gather + All-reduce
- ☐ All-reduce = All-gather + Broadcast
- ✓ ☒ All-reduce = Reduce-scatter + All-gather

Done